It turns out that the theory of representations of quivers is a part of the theory of representations of algebras in the sense that for each quiver Q, there exists a certain algebra P_Q, called the path algebra of Q, such that a representation of the quiver Q is "the same" as a representation of the algebra P_Q. We shall first define the path algebra of a quiver and then justify our claim that representations of these two objects are "the same".

Definition 2.8.4. The **path algebra** P_Q of a quiver Q is the algebra whose basis is formed by oriented paths in Q, including the trivial paths p_i, $i \in I$, corresponding to the vertices of Q, and multiplication is the concatenation of paths: ab is the path obtained by first tracing b and then a. If two paths cannot be concatenated, the product is defined to be zero.[2]

Remark 2.8.5. It is easy to see that if Q is a finite set then $\sum\limits_{i \in I} p_i = 1$, so P_Q is an algebra with unit.

Problem 2.8.6. Show that for a finite quiver Q the algebra P_Q is generated by p_i for $i \in I$ and a_h for $h \in E$ with the following defining relations:

(1) $\sum_{i \in I} p_i = 1$,
(2) $p_i^2 = p_i$, $p_i p_j = 0$ for $i \neq j$,
(3) $a_h p_{h'} = a_h$, $a_h p_j = 0$ for $j \neq h'$,
(4) $p_{h''} a_h = a_h$, $p_i a_h = 0$ for $i \neq h''$.

We now justify our statement that a representation of a quiver is the same thing as a representation of the path algebra of a quiver.

Let $\mathbf{V}$ be a representation of the path algebra P_Q. From this representation, we can construct a representation of Q as follows: let $V_i = p_i \mathbf{V}$, and for any edge h, let $x_h = a_h|_{p_{h'}\mathbf{V}} : p_{h'}\mathbf{V} \longrightarrow p_{h''}\mathbf{V}$ be the operator corresponding to the one-edge path h.

Similarly, let (V_i, x_h) be a representation of a quiver Q. From this representation, we can construct a representation of the path algebra

[2] An oriented path is specified by a nonnegative integer n, a sequence of vertices $i_0, \ldots, i_n$, and a sequence of edges $e_1, \ldots, e_n$ such that each e_r has source i_{r-1} and target i_r. In particular, when $n = 0$, one still must choose one vertex i_0, which explains why there is one p_i for each $i \in I$. Two paths a, b can be concatenated to form the path ab if and only if the final target of a equals the first source of b.